

CSCE 676 DATA MINING · MODULE 13 · LECTURE NOTES

Claiming a Pattern Honestly

What a discovery establishes.

MODULE OUTCOMES

- State what a found pattern establishes.
- Identify the standard overclaims.
- Write a defensible discovery claim.
- Place this course relative to CSCE 638 and CSCE 670.
- State a proportionate practice for real mining work.

1 What a finding establishes

A FOUND PATTERN IS A STATEMENT ABOUT YOUR SEARCH PROCEDURE UNTIL IT HAS BEEN VALIDATED

It establishes that this statistic, computed on this data, after this search, exceeded this threshold — all four clauses, and the third is the one usually omitted entirely.

And without the hypothesis count (Module 11 §1), the significance simply cannot be interpreted — because the same p-value means entirely different things after one test and after a million, and the reader has no way to tell which happened.

With validation on untouched held-out data, it becomes a statement about the data rather than about your procedure — which is the qualitative change, and is why Module 11 §4's third level matters so much more than the second.

And it is still not causal (Module 11 §4's closing callout) — which requires something the observational data cannot supply, and saying specifically what is the honest form rather than appending a generic disclaimer.

The claim	The correction
'We discovered that X causes Y.'	You found an association. Name the experiment you would need to make it causal (Module 11 §4).
'Significant at $p < 0.001$.'	Out of how many hypotheses tested? (Module 11 §1.) Without that, the number is uninterpretable.
'The data has four clusters.'	Your algorithm returned four, because you asked for four. What does a null comparison score? (Module 04 §4.)

The claim	The correction
'The network has strong community structure.'	What is the modularity against a degree-preserving null model? (Module 07 §3.)
'Our detector is 99% accurate.'	What is the precision at a reviewable alert volume? (Module 08 §3.) Accuracy is meaningless at a low base rate.
'The data is anonymised.'	Not establishable by inspecting your own dataset (Module 12 §1).

Every row substitutes an algorithm's output for a property of the world — four clusters returned for four clusters existing, a modularity score for community structure, an accuracy figure for usefulness. Which is this course's single characteristic error, and it is the thing Module 01 §1 set up by separating the two problems. The shared structure across the rows is what makes the table worth memorising rather than the individual corrections.

2 Claims you can defend

- 'Among 4.2 million tested pairs, 118 exceeded our threshold after Benjamini–Hochberg correction at 5%; of those, 31 replicated on held-out data we had not examined.' The search is disclosed, the correction is named, and the replication rate is the headline.
- 'The effect is a 2.1% difference, with a 95% confidence interval of 1.8% to 2.4% — which exceeds the 1% we specified in advance as the threshold for materiality.' (Module 11 §3, and Module 01 §4's fourth item.)
- 'Our minhash estimates had a mean absolute error of 0.031 against exact Jaccard on 10,000 sampled pairs, which is consistent with the 0.035 bound at $k = 200$.' The approximation verified rather than assumed (Module 02 §4).
- 'Modularity is 0.71; a degree-preserving null model achieves 0.42 with a standard deviation of 0.01.' Which is what makes the 0.71 interpretable, and without the second clause it is not (Module 07 §3).
- And: 'we cannot distinguish this pattern from a selection effect in how the data was collected.' The last one is the limitation statement, and it is Project 2's hardest-graded requirement — because it requires knowing what your design cannot settle, which is harder than knowing what it did.

3 The semester

Course	What it covers	The proxy it has to live with
CSCE 638	Language as input and output.	A metric standing in for a quality judgement — BLEU for a good translation, ROUGE for a good summary.
CSCE 676 (this one)	Finding structure in data at scale.	A score standing in for a pattern being real — modularity for community structure, silhouette for clusters, support for an association.
CSCE 670	Satisfying an information need.	A relevance label standing in for relevance to an actual person with an actual need.

All three are courses about a proxy, and in all three the discipline is the same: say what the proxy is, and say what it does not capture. Which makes the choose-the-metric-from-the-use argument (CSCE 633 Module 12) the semester's organising idea, arriving in three settings with three different proxies and one shared requirement. The proxy framing unifies the semester, and it is worth carrying forward.

4 A proportionate practice

- 1 · Split the data before you look at it, and keep the holdout genuinely untouched (Module 01 §4) — **free, and the largest single improvement available in Module 11 §4's hierarchy of evidence.**
- 2 · Keep the hypothesis log, including the abandoned branches (Module 11 §1) — **cheap, and nothing downstream works without it**, because no correction can be applied to an unknown count.
- 3 · Report effect sizes with confidence intervals, and decide in advance what magnitude would matter (Module 11 §3) — which prevents the significant-and-useless finding that large n guarantees.
- 4 · Run the null comparison for any unsupervised finding — clusters (Module 04 §3), communities (Module 07 §3), anomalies (Module 08 §4) — **which is the only way any of those scores can be read.**
- 5 · And verify your approximation bounds on your own data (Module 01 §2, Module 02 §4, Module 06 §4), **which is the other half of the course** and the half that makes the algorithms trustworthy. **The first two cost essentially nothing and change what your findings are worth — which makes omitting them quite hard to defend.**

WHERE THIS COURSE LEAVES YOU

You can find structure in data too large to inspect, with a stated approximation bound — similar items, frequent patterns, clusters, low-dimensional structure, stream summaries, communities, and anomalies — **and you verify the bound rather than citing it.**

You know what scale does to the cost model — passes first, then shuffles, and never arithmetic — and **you check whether a single machine is the right answer** before provisioning a cluster (Module 10 §3).

And you know that searching hard enough guarantees a finding — so **you count the hypotheses, apply the correction, validate on held-out data, and report the effect size rather than the p-value**, which together are what distinguish a result from an artefact.

The closing rule is the program's, unchanged across twenty-nine courses: state what you measured, state what you assumed, and never claim more than you established. In this subject it means saying how hard you looked — because **a pattern reported without its search is uninterpretable**, and §1's table is what happens when that is left out.

Where to see this taught differently

Original material only goes so far — hearing the same idea in another voice is how it sticks. These are free, and they cover this module specifically.

Resource	What it covers for this module
Mining of Massive Datasets, chapter 1 (free)	§1's position, from the authors — reread the bonferroni's-principle section now that you have Module 11.
Ioannidis, and the reproducibility literature (free)	§1 and §4's motivation — the third reading, and it reads differently each time.
Dacrema et al. — Are We Really Making Much Progress? (free)	§2's baseline requirement, demonstrated — and Module 09 §4's finding, which generalises.
Leek & Peng — Statistics: P values are just the tip of the iceberg (free)	§4's ordering — why the design decisions matter more than the statistical test.

Exercises

Do these in code, not on paper, unless the exercise says otherwise. The module is not finished until they run.

- 1 **Take a published data mining finding** and assess it against §1's table.
- 2 **Determine whether its hypothesis count is reported.**
- 3 **Rewrite one of your own findings** in §2's form, with the search disclosed.
- 4 **Write the limitation statement** for it.

- 5 **Identify the proxy** in each of the three Semester 10 courses.
- 6 **Audit your own practice** against §4's five habits and report the gaps.
- 7 **Check whether your holdout is genuinely untouched.**
- 8 **Count the entries in your hypothesis log** and compare against what you would have guessed in Module 01.
- 9 **Revisit Module 01's last exercise** and compare your predictions.
- 10 **Project 2 is now due.** Submit the pattern with its effect size, the hypothesis count and correction, the held-out validation, the three ruled-out alternatives, the causal statement, and the privacy assessment.

Self-check

Answer these in writing, from memory, without looking back at the notes. If you cannot, you have read this module but not yet learned it — that is useful information, not a failure.

- 1 What four clauses does a found pattern's claim require?
- 2 Why can significance not be interpreted without the hypothesis count?
- 3 What changes when a pattern survives untouched held-out data?
- 4 Give six overclaims and the correction to each.
- 5 What do all six have in common?
- 6 Give four defensible claim forms.
- 7 What is Project 2's hardest requirement, and why?
- 8 Name the proxy in each of the three Semester 10 courses.
- 9 Give the five habits in order, and which two are nearly free.
- 10 In this course, what does 'never claim more than you established' mean?